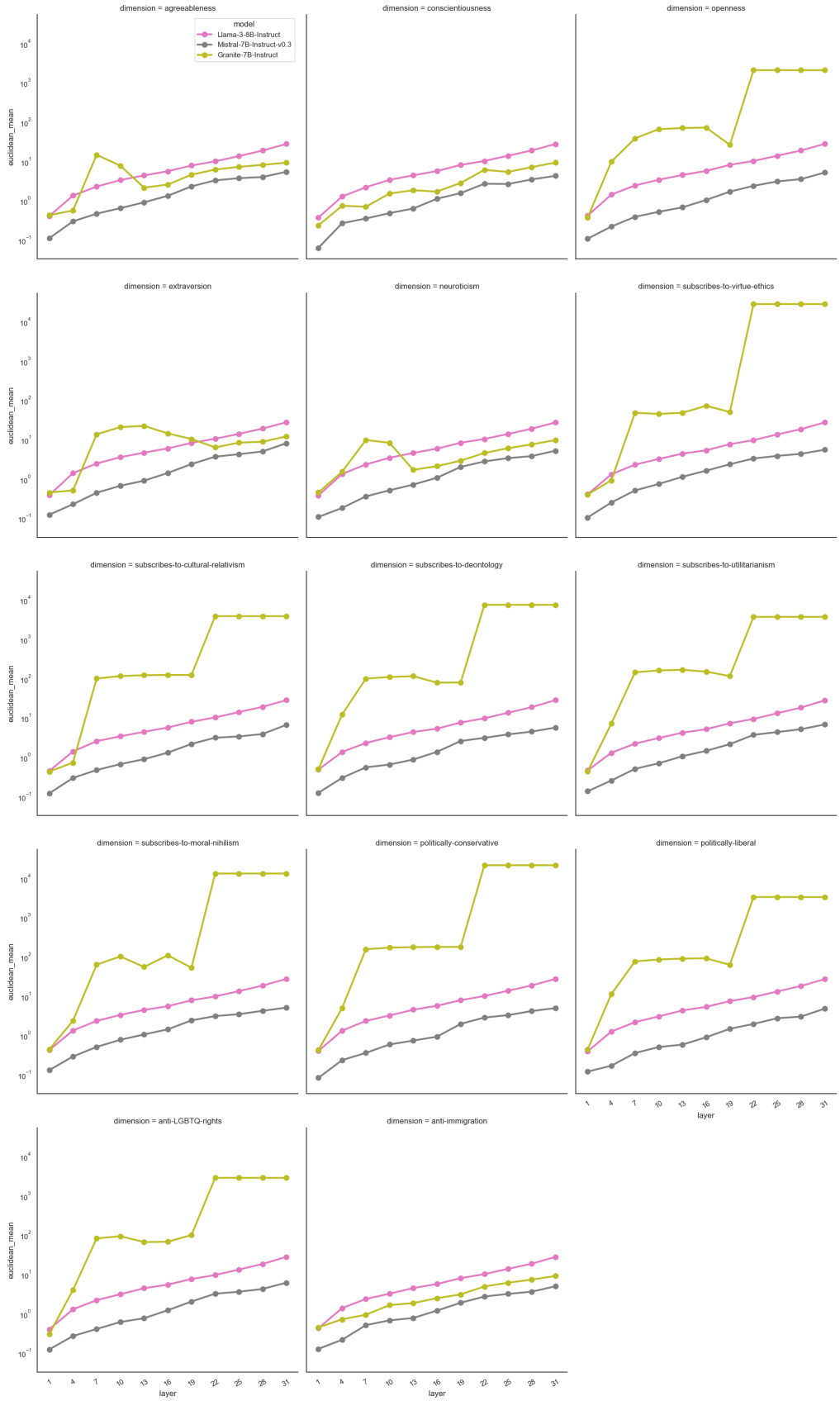

Figure 6: **(Q1)**: Euclidean distances stratified by dimensions across models and layers.

Table 4: **(Q1)**: Validation of usefulness of the set of salient activations O_{S^*} at *Layer 1* and *31* of *Llama3-8B-Instruct* .

Dimension	Layer	$\lvert O_{S^*}\rvert$	Precision (↑)	Recall (↑)
AGREE	1	2815	0.5067 ± 0.1432	0.2469 ± 0.0904
	31	1167	$\mathbf{0.9971 \pm 0.0113}$	$\mathbf{0.9979 \pm 0.00984}$
CONSC	1	3240	0.3895 ± 0.1753	0.2237 ± 0.10208
	31	1210	$\mathbf{0.9992 \pm 0.0001}$	$\mathbf{0.95454 \pm 0.04876}$
OPEN	1	3204	0.6048 ± 0.2317	0.2434 ± 0.10216
	31	1177	$\mathbf{0.9998 \pm 0.0003}$	$\mathbf{0.97727 \pm 0.0422}$

A.3 Extended Results

A.3.1 (Q1) Persona Representations Across Layers in Other Models

In Tab. 4, we observe low precision and recall in early layers. This suggests that the activation locations found are not useful to determine MATCHINGBEHAVIOR for dimensions, compared to performance in *Layer 31*. In *Layer 31*, we observe high precision and recall.

This also confirms the quality of the representations that we observe in Fig. 7 and 8, where we plot the PCA embeddings for the *openness* persona in *Granite-7B-Instruct* and the *conscientiousness* persona in *Mistral-7B-Instruct*, respectively. We observe that the separability between MATCHINGBEHAVIOR and NOTMATCHINGBEHAVIOR representations improves in the later layers.

In Tables 5, 6, and 7, we show several clustering metrics to quantify the separation between q^+ (MATCHINGBEHAVIOR) representations and q^- (NOTMATCHINGBEHAVIOR).

A.3.2 (Q2) Unique Locations of Persona Within a Layer

Fig. 9, shows Upset and Venn diagrams plots for intra-persona (Level 2) analysis for personas from all topics, *Personality*, *Ethics*, *Politics*.

In Tab. 8 we report precision and recall regarding MATCHINGBEHAVIOR and NOTMATCHINGBEHAVIOR detection (Level 2) comparing different unsupervised methods. We observe that Deep Scan outperforms in precision while still maintaining a high recall compared to the other unsupervised methods.

In Fig. 10, we show a Venn diagram of the overlap of salient activations at the inter-topic level (Level 0). Between personas from *Ethics*, *Politics*, and *Personality*, we observe very low overlap between salient activations.

[12] https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct
[13] https://huggingface.co/ibm-granite/granite-7b-instruct
[14] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3

Table 5: **(Q1)**: Metrics to assess the goodness of *layer 1* and *31* to detect personas (Level 2) in *Llama3-8B-Instruct*. Metrics: Silhouette Score (SH), Calinski-Harabasz Score (CH), Euclidean Distance (ED), Davies-Bouldin Score (DB).

Model	Dimension	layer	SH (↑)	CH (↑)	DB (↓)	ED (↑)
Llama3-8B-Instruct	AGREE	1	0.500 ± 0.0741	340.6 ± 163.9	0.731 ± 0.072	0.403 ± 0.012
		31	0.792 ± 0.0000	3264.5 ± 0.002	0.326 ± 0.000	27.57 ± 0.000
	CONSC	1	0.635 ± 0.0000	718.8 ± 0.026	0.569 ± 0.000	0.370 ± 0.000
		31	0.813 ± 0.0000	4150.4 ± 0.003	0.285 ± 0.000	27.47 ± 0.000
	OPEN	1	0.602 ± 0.0000	570.2 ± 0.0005	0.645 ± 0.000	0.414 ± 0.000
		31	0.795 ± 0.0000	3564.1 ± 0.0125	0.319 ± 0.000	27.60 ± 0.000
	EXTRA	1	0.578 ± 0.0000	527.5 ± 0.001	0.705 ± 0.000	0.382 ± 0.000
		31	0.788 ± 0.0000	3176.5 ± 0.001	0.330 ± 0.000	27.47 ± 0.000
	NEURO	1	0.584 ± 0.0000	615.0 ± 0.065	0.686 ± 0.000	0.378 ± 0.000
		31	0.796 ± 0.0000	3372.4 ± 0.001	0.306 ± 0.000	27.22 ± 0.000
	VIRTUE	1	0.614 ± 0.0000	644.7 ± 0.007	0.639 ± 0.000	0.402 ± 0.000
		31	0.797 ± 0.0000	3471.5 ± 0.000	0.309 ± 0.000	27.24 ± 0.000
	RELAT	1	0.500 ± 0.0000	405.4 ± 0.000	0.935 ± 0.000	0.440 ± 0.000
		31	0.814 ± 0.0000	3960.0 ± 0.002	0.276 ± 0.000	28.06 ± 0.000
	DEONT	1	0.523 ± 0.0230	315.1 ± 104.9	0.758 ± 0.000	0.479 ± 0.054
		31	0.824 ± 0.0000	4297.7 ± 0.004	0.254 ± 0.000	28.17 ± 0.000
	UTILI	1	0.542 ± 0.0405	326.0 ± 160.5	0.645 ± 0.000	0.454 ± 0.062
		31	0.798 ± 0.0000	3587.6 ± 0.001	0.301 ± 0.000	27.59 ± 0.000
	NIHIL	1	0.556 ± 0.0000	472.8 ± 0.001	0.810 ± 0.000	0.421 ± 0.000
		31	0.808 ± 0.0000	3909.8 ± 0.016	0.282 ± 0.000	27.20 ± 0.000
	CONS	1	0.685 ± 0.0001	860.7 ± 0.024	0.475 ± 0.000	0.400 ± 0.000
		31	0.818 ± 0.0000	4529.9 ± 0.007	0.266 ± 0.000	27.17 ± 0.000
	LIBER	1	0.692 ± 0.0001	897.2 ± 0.023	0.469 ± 0.000	0.387 ± 0.000
		31	0.803 ± 0.0000	3809.3 ± 0.004	0.294 ± 0.000	26.80 ± 0.000
	LGBTQ	1	0.634 ± 0.0000	703.1 ± 0.014	0.573 ± 0.000	0.399 ± 0.000
		31	0.774 ± 0.0000	2815.3 ± 0.034	0.352 ± 0.000	27.62 ± 0.000
	IMMI	1	0.570 ± 0.0000	476.5 ± 0.000	0.730 ± 0.000	0.424 ± 0.000
		31	0.806 ± 0.0000	3756.4 ± 0.002	0.288 ± 0.000	27.67 ± 0.000